# Michael Ichikawa

AI ENGINEER · AGENTIC SYSTEMS

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## SUMMARY

I shipped a production AI agent solo, from an empty repo to a listing on the Microsoft Commercial Marketplace and the Teams Store, doing the product, the engineering and the compliance myself. I built a multi-agent workflow engine from scratch for the same class of problem a level up, composing single-purpose agents into a validated graph, with a human approval gate before any step that acts. Before all of it, MS Mathematics, eight years teaching undergraduate math, and three years of semiconductor physical design at Intel.

## FLAGSHIP WORK

### Microclaw — Production AI Agent for Microsoft 365

Microsoft Marketplace listing

One agent across Microsoft 365, with 87 function-calling tools and a tool-calling loop that chains them until a task is complete. The hard part was choosing: sending every tool on every message is expensive, so a retrieval step narrows the list first, and I measure how often that drops a tool the request needed. Around that: RAG semantic memory, model routing that escalates to a larger model only when a turn needs it, and multi-layer failure recovery.

*TypeScript · Azure OpenAI · Microsoft Graph · Azure Postgres (RLS) · Bicep*

### Multi-Agent Workflow Engine — built from scratch

mtichikawa.github.io/workflow-engine

A Python engine, built from scratch on the standard library, that turns a plain-English request into a graph of single-purpose agents. Every workflow is statically validated before it runs, and any step that acts outside the engine must sit behind a human approval gate. A learning loop infers a team's unwritten rules from their corrections, with no fine-tuning: on 150 blind-written messages, routing accuracy rose from 59 to 97 percent. On 320 blind-written patient messages, its triage step sent all 160 medical ones to a person, with no false alarms.

*Python · agentic orchestration · MCP (client + server) · contracts + recipes · static graph validation*

## TECHNICAL SKILLS

| | |
|---|---|
| AI-Native Dev | directing AI agents to implement, test and debug complex features · Claude Code (daily driver) · agent-directed builds verified by tests and evals · per-repo context files · AI-native development practices |
| Agents / LLM | agent orchestration · tool-calling agent loops · function calling · tool use · tool retrieval (KNN) · RAG (retrieval-augmented generation) · vector search & embeddings (pgvector · FAISS · BigQuery Vector Search) · semantic memory · multi-agent workflows · human-in-the-loop gating · MCP (client + server) · in-context learning |
| Reliability & Cost | failure-mode engineering (backoff/retry · context compaction · loop detection · abort ceilings) · model routing with mid-task escalation · input-token reduction · multi-tenant isolation (Postgres RLS) |
| Evaluation | eval design · expected-facts scoring · A/B against baselines · blind held-out test sets · pytest · Vitest · GitHub Actions CI |
| Built with | Python · TypeScript · Node.js · SQL · Azure OpenAI · Anthropic API · Microsoft Graph · PostgreSQL · BigQuery · Delta Lake (medallion) · Spark Structured Streaming · SQLAlchemy · pandas · scikit-learn · HuggingFace · sentence-transformers · Azure (Container Apps · Entra · Bicep) · AWS (S3 · Lambda · DynamoDB) · GCP (Vertex AI · Cloud Run) · Docker · FastAPI / REST · Redis |

## EXPERIENCE

### Founding AI/ML Engineer

2026 – present

Microclaw LLC · Portland, OR

- Built the agent loop and redesigned its tool selection: on 362 blind-written test requests it offered every tool a request needed 98.1 percent of the time, up from 87.0, and on a 142-case agent eval it called the right tools on 134 of 142 cases on average over three runs, with 49 percent fewer prompt tokens than sending every tool
- About ten weeks from first commit to a Marketplace listing, solo, through Microsoft Marketplace certification and the Fulfillment and Metering API work that makes it billable, then Teams Store validation after seven review rounds
- Multi-tenant isolation via Postgres row-level security: FORCE policies on every tenant-scoped table, an app role that cannot bypass them, and delegated Entra auth

### Career Break — Full-Time Parent · Self-Directed Transition to Software

2023 – 2026

Portland, OR

- Three years as primary caregiver, retraining into software throughout
- Started in data engineering and ML, then moved to agentic development as the field did

### Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught mathematics for eight years at an open-enrollment college: Algebra through Calculus III, Linear Algebra, Differential Equations and introductory Statistics
- Wrote my own lectures and exams, found the topics students kept missing, and reworked how I taught them
- Wrote my own version of an open-source precalculus textbook in LaTeX, reorganized by concept so trig functions came in week 2 instead of week 10

### Physical Design Engineer

2008 – 2011

Intel Corporation · Semiconductor R&D

- Owned the layout of whole blocks in the Ivy Bridge (22nm) microprocessor, closing timing, power and design-rule checks
- Kept my blocks passing every check as cell-library updates and upstream circuit changes landed in a large, version-controlled shared design
- Wrote Python and Perl automation for physical verification

## EDUCATION

**MS Mathematics** — The City College of New York — 2014

**BS Mechanical Engineering** — University of California, Berkeley — 2008

## PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Self-directed work from the 2023–2026 transition above, before the move to agentic systems. The one closest to current work is a **RAG pipeline on GCP** with three interchangeable retrievers behind one interface, BigQuery Vector Search, FAISS and pgvector, so they could be compared on the same eval.

The rest spans data engineering and ML: streaming and lakehouse pipelines · cloud ETL · anomaly detection · LLM tooling and bias testing · forecasting · a trading backtest system end to end.